

UNIVERSITY OF MAURITIUS
FACULTY OF INFORMATION, COMMUNICATION
AND DIGITAL TECHNOLOGIES



SPECIAL RETAKE EXAMINATIONS

AUGUST 2017

PROGRAMME	BSc (Hons) Computer Applications		
MODULE NAME	Information Systems Security		
DATE	Wednesday 16 August 2017	MODULE CODE	CSE 3207(5)
TIME	09:30 – 11:30 Hrs	DURATION	2 Hours
NO. OF QUESTIONS SET	4	NO. OF QUESTIONS TO BE ATTEMPTED	4

INSTRUCTIONS TO CANDIDATES

Answer all questions.

All questions carry equal marks.

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Question 1

- (a) Explain the importance of the OSI Security Architecture framework and describe briefly its content.
[5 marks]
- (b) Explain the difference between 'active' and 'passive' security threats.
[4 marks]
- (c) Briefly define three categories of security services.
[2 x 3 marks]
- (d) By making use of a matrix show the relationship between security services and attacks.
[5 marks]
- (e) A security policy is a preventative mechanism for protecting important data and processes of a company. List five security measures which a company can apply for keeping its physical interfaces secured.
[5 marks]

Question 2

- (a) A security risk survey conducted at a particular organisation valued its asset at Rs 25 million. Potential threat identified at the company has been estimated to have a risk exposure of 6% with rate of occurrence of only once every two years. Calculate the highest amount that the company needs to invest annually as a countermeasure for protecting its computing resources. **[5 marks]**
- (b) Describe the Vigenère Cipher encryption technique. **[5 marks]**
- (c) The one time pad system can be used to enforce security from a message sender to the right receiver. Explain the problems that a user may face when using the one time pad system. **[5 marks]**
- (d) Apply the Play Fair Cipher to encrypt the message below using the keyword 'SPECIAL'
- 'HACKERS MAKE THE SYSTEM VULNERABLE'
- [6 marks]**
- (e) Explain the difference between an unconditionally secure cipher and a computationally secure cipher. **[4 marks]**

Question 3

- (a) Explain the difficulties that are encountered when using symmetric key data encryption techniques.
[5 marks]
- (b) Differentiate between 'stream' and 'block' ciphers.
[5 marks]
- (c) Explain why some block cipher modes of operation only use encryption, while others use both encryption and decryption.
[5 marks]
- (d) By making use of a block diagram, explain the different rounds of the Feistel Cipher Network.
[5 marks]
- (e) Explain the features of Double DES.
[5 marks]

Question 4

- (a) Explain what is a 'meet-in-the-middle' attack.
[5 marks]
- (b) In a symmetric key cryptography system, determine the number of secret keys needed if in a team of 10 people, every member needs to communicate with every other member
[5 marks]
- (c) Given that $p=3$, $q=11$, $e=7$, $m=5$, use the RSA algorithm to find the public and private keys and the ciphertext C .
[9 marks]
- (d) Describe the functionality of Steganography.
[3 marks]
- (e) Describe the three main processes involved in water marking.
[3 marks]

END OF QUESTION PAPER